



Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. For a given base VA and base voltage, the PU impedance of a power system element is Z. What will be the PU impedance of this element when both voltage and VA bases are doubled?
A) Z
B) 2Z
C) 0.5 Z
D) 0.25 Z
2. The per unit impedance of 200 kVA, 13.2 kV synchronous motor having a reactance of 50 Ω per phase is
A) 0.7575 PU
B) 57.39 PU
C) 3.3 PU
D) 0.0574 PU
3. If X_a is the armature reaction reactance, X_s is the synchronous reactance and X_l is the leakage reactance of an alternator, then which of the following relations is true?
A) $X_a = X_l = X_s$
B) $X_a = X_l + X_s$
C) $X_a + X_l = X_s$
D) $X_l = X_a + X_s$
4. Which of the following is retained during formation of reactance diagram from impedance diagram?
A) Resistance of transmission lines
B) Shunt admittance of transmission lines
C) Exciting branch of transformers
D) Series reactance of transmission lines
5. The steady state fault current during a three-phase terminal fault on a generator is limited by
A) Synchronous reactance of generator
B) Transient reactance of generator
C) Sub-transient reactance of generator
D) DC component in the fault current
6. If μ represents the instant on voltage waveform at which symmetrical short circuit occurs in an R-L circuit and δ is the phase angle, then the transient dc component of fault current is absent
A) When $\mu = 90^\circ$ and $\delta = 0^\circ$
B) When $\mu = 0^\circ$ and $\delta = 90^\circ$
C) When $\mu = 45^\circ$ and $\delta = 0^\circ$
D) When $\mu = \delta$
7. As applied to a symmetrical fault, which among the following statements is true?
A) Sub-transient period is shorter than transient period
B) Sub-transient and transient periods last equally long
C) Transient period is shorter than sub-transient period
D) Steady-state period is shorter than transient period
8. A load can be considered symmetrical only if
A) Self-impedances of three phases are different and mutual impedances are unequal
B) Self-impedances of three phases are same and mutual impedances are equal
C) Self-impedances of three phases are same and mutual impedances are unequal
D) Self-impedances of three phases are unequal and mutual impedances are same

9. The sequence networks are decoupled
 - A) Only when the source voltage is balanced and load impedance is symmetrical
 - B) Even when the source voltage is unbalanced provided load impedance is symmetrical
 - C) When the source voltage is unbalanced and load impedance is not symmetrical
 - D) When the source voltage is balanced and load impedance is not symmetrical
10. The neutral current is
 - A) Equal to zero sequence current
 - B) Equal to one third the zero sequence current
 - C) Equal to three times the zero sequence current
 - D) Equal to six times the zero sequence current
11. In a three-phase star-connected four-wire system, zero sequence components are absent in
 - A) Phase currents
 - B) Line currents
 - C) Line voltages
 - D) Phase voltages
12. In which of the following three-phase transformer connection configurations, there can be zero sequence currents on the input lines?
 - A) Grounded star / delta
 - B) Ungrounded star / grounded star
 - C) Ungrounded star / delta
 - D) Ungrounded star / ungrounded star
13. The zero sequence current is absent in case of
 - A) L-G fault
 - B) L-L fault
 - C) L-L-G fault
 - D) L-L-L-G fault
14. The sequence components of voltage are equal in case of
 - A) L-G fault
 - B) L-L fault
 - C) L-L-G fault
 - D) L-L-L-G fault
15. For determining the fault current, the three sequence networks have to be connected in series
 - A) For L-G fault
 - B) For L-L fault
 - C) For L-L-G fault
 - D) For L-L-L-G fault
16. An unloaded star-connected alternator with an ungrounded neutral is subjected to a shunt fault, but the fault current is found to be zero. The type of the fault is:
 - A) L-G
 - B) L-L
 - C) L-L-G
 - D) L-L-L-G
17. Reluctance power is present
 - A) Only in non-salient-pole alternator
 - B) Only on salient-pole alternator
 - C) Both in salient- and non-salient pole alternators
 - D) Neither in salient nor in non-salient pole alternators
18. Steady-state stability limit of an alternator
 - A) Is lower than its transient stability limit
 - B) Is higher than its transient stability limit
 - C) Is same as its transient stability limit
 - D) May be higher or lower than its transient stability limit
19. The angle δ in the swing equation of a synchronous generator is the
 - A) angle between stator voltage and current
 - B) angular displacement of the rotor with respect to the stator
 - C) angular displacement of the stator mmf with respect to a synchronously rotating axis
 - D) angular displacement of a rotor axis with respect to a synchronously rotating axis
20. The swing equation is a
 - A) Second order linear differential equation
 - B) First order linear differential equation
 - C) First order nonlinear differential equation
 - D) Second order nonlinear differential equation

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) What is per unit? Mention the advantages of per unit quantities. | 8 | L*2 | 1 | 1 |
| b) A synchronous generator rated 500 kVA, 440 V, 0.1 p.u. subtransient reactance is supplying a passive load of 400 kW at 0.8 p.f. lag. Calculate the initial symmetrical current (rms) for a fault at the generator terminals. Take 500 kVA and 440 V as the base values for the generator. | 8 | L2 | 2 | 1,2 |
| 2. a) Define short-circuit MVA of a system. Describe its significance. | 6 | L1 | 2 | 1 |
| b) A 100 MVA, 33 kV, 3-phase generator has a sub-transient reactance of 15%. The generator is connected to three motors through a transmission line and transformers as shown in Fig. Q2(b). The motors are rated for 30 MVA, 20 MVA and 50 MVA at 30 kV with 20% subtransient reactance each. The transformers are rated 32 kV Delta / 100 kV Star, 100 MVA with leakage reactance of 8%. The line has a reactance of 50 Ω . Selecting the generator ratings as the base in the generator circuit, draw the per unit reactance diagram. | | | | |

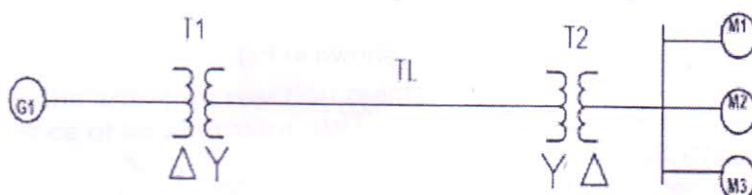


Fig. Q2(b)

- | | | | | |
|--|----|----|---|---|
| 3. a) Fig. Q. 3(a) shows the reactance diagram of a radial transmission system. The ratings and reactances of various components are as shown therein. A load of 60 MW at 0.9 p.f. lagging is tapped from the 66 kV substation which is to be maintained at 60 kV. Calculate the terminal voltage of the machine. Represent the transmission line and transformer by series reactance only. Take base values as 100 MVA and 220 kV on the transmission line. | 10 | L3 | 1 | 1 |
|--|----|----|---|---|

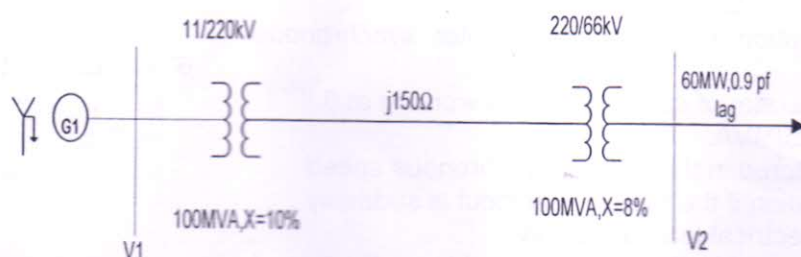


Fig. Q. 3(a)

- | | | | | |
|---|----|----|---|-----|
| b) With relevant equations, elaborate doubling effect in a series R-L circuit subjected to a symmetrical fault. | 10 | L3 | 1 | 1,2 |
| | 6 | L2 | 2 | 1 |

Unit – II

4. a) Show that in case of a symmetrical three-phase load, current of a particular sequence produces voltage drop of the same sequence only. 8 L2 3 1
- b) A salient-pole generator without dampers is rated 20 MVA, 13.8 kV and has a sub-transient reactance of 0.25 p.u. The negative and zero sequence reactances are 0.35 p.u. and 0.1 p.u. respectively. The neutral of the generator is solidly grounded. Calculate the fault current in Amperes when a single line-to-ground fault occurs at the terminals of the generator. Assume pre-fault current is zero. Take generator ratings as the base values. 8 L3 4 1
5. a) Draw the zero sequence diagrams for the following configurations of three-phase transformers:
 (i) Grounded star/ ungrounded star
 (ii) Grounded star / grounded star
 (iii) Grounded star / delta
 (iv) Delta / delta.
 Please take the grounding impedance as zero. 8 L2 3 1
- b) Derive an expression for fault current for a line-line fault on an unloaded generator. 8 L2 4 1
6. a) Obtain an expression for complex power in a three-phase system in terms of sequence components of voltages and currents. 6 L2 3 1
- b) The following data may be assumed for the network shown in Fig. Q6(b).
 Generator: 50 MVA, 11 kV, $X_1 = 80\%$, $X_2 = 50\%$, $X_0 = 20\%$.
 Transformer: 40 MVA, 11/110 kV, $X_1 = X_2 = X_0 = 6\%$.
 The percentage reactances are given on the base of ratings of each device. If a double line-ground fault occurs at F, find the conductor current in Amperes at the point of fault. Take 50 MVA and 11 kV as the base values for the generator circuit.

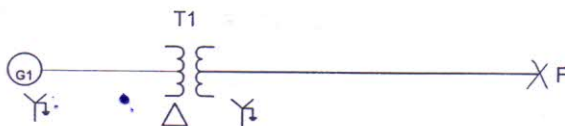


Fig. Q6(b).

10 L3 4 1,2

Unit – III

7. a) Derive power-angle equation for a cylindrical rotor synchronous machine. 6 L2 5 1
- b) A 6-pole, 50 Hz turbogenerator of capacity 80 MW working at 0.8 p.f. has an inertia of 10 MJ/MVA.
 (i) Calculate the energy stored in the rotor at synchronous speed.
 (ii) Find the rotor acceleration if the mechanical input is suddenly raised to 75 MW for an electrical load of 60 MW.
 (iii) Supposing the above acceleration is maintained for a duration of 6 cycles, calculate the change in torque angle. 10 L3 5 1,2
8. a) Discuss equal area criterion in the context of transient stability. 8 L3 5 1,2
- b) Delineate the steps involved in point-by-point method of solution of swing equation. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
